

Non-celiac wheat sensitivity diagnosed by double-blind placebo-controlled challenge: exploring a new clinical entity.

OBJECTIVES: Non-celiac wheat sensitivity (WS) is considered a new clinical entity. An increasing percentage of the general population avoids gluten ingestion. However, the real existence of this condition is debated and specific markers are lacking. Our aim was thus to demonstrate the existence of WS and define its clinical, serologic, and histological markers. **METHODS:** We reviewed the clinical charts of all subjects with an irritable bowel syndrome (IBS)-like presentation who had been diagnosed with WS using a double-blind placebo-controlled (DBPC) challenge in the years 2001-2011. One hundred celiac disease (CD) patients and fifty IBS patients served as controls. **RESULTS:** Two hundred and seventy-six patients with WS, as diagnosed by DBPC challenge, were included. Two groups showing distinct clinical characteristics were identified: WS alone (group 1) and WS associated with multiple food hypersensitivity (group 2). As a whole group, the WS patients showed a higher frequency of anemia, weight loss, self-reported wheat intolerance, coexistent atopy, and food allergy in infancy than the IBS controls. There was also a higher frequency of positive serum assays for IgG/IgA anti-gliadin and cytometric basophil activation in "in vitro" assay. The main histology characteristic of WS patients was eosinophil infiltration of the duodenal and colon mucosa. Patients with WS alone were characterized by clinical features very similar to those found in CD patients. Patients with multiple food sensitivity were characterized by clinical features similar to those found in allergic patients. **CONCLUSIONS:** Our data confirm the existence of non-celiac WS as a distinct clinical condition. We also suggest the existence of two distinct populations of subjects with WS: one with characteristics more similar to CD and the other with characteristics pointing to food allergy.